

1. Random Variables X, Y on $(\Omega, \mathcal{F}, P)$ and $A \in \mathcal{F}$
 Show that $Z: \Omega \rightarrow \mathbb{R}$, $Z(\omega) = X(\omega)I_A(\omega) + Y(\omega)I_{A^c}(\omega)$

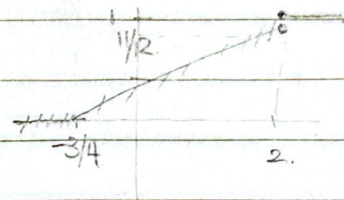
$$Z(\omega) = \begin{cases} X(\omega) & \omega \in A \\ Y(\omega) & \omega \notin A \end{cases}$$

$$\therefore Z^{\leftarrow}(B) = \{\omega \mid Z(\omega) \in B\} = (A \cap \{\omega \mid X(\omega) \in B\}) \cup (A^c \cap \{\omega \mid Y(\omega) \in B\})$$

$$A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}, X^{\leftarrow}(B) \in \mathcal{F}, Y^{\leftarrow}(B) \in \mathcal{F}$$

$$\therefore Z^{\leftarrow}(B) \in \mathcal{F} \Rightarrow Z \text{ is random variable}$$

2.
$$F_X(x) = \begin{cases} 0 & x < -3/4 \\ 1/3x + 1/4 & -3/4 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$



$$Y = 2X^2$$

$$\begin{aligned} \text{a) } P(1/2 < X < 3/2) &= P(1/2 < X \leq 3/2) - P(X = 3/2) \\ &= F_X(3/2) - F_X(1/2) - \left\{ F_X(3/2) - \lim_{x \rightarrow 3/2^-} F_X(x) \right\} \\ &= 3/4 - 5/2 - (3/4 - 3/4) = 1/3. \end{aligned}$$

$$\begin{aligned} \text{b) } P(X+Y \leq 2) &= P(X+2X^2 \leq 2) = P\left(\frac{-1-\sqrt{17}}{4} \leq X \leq \frac{-1+\sqrt{17}}{4}\right) \\ &= P\left(\frac{-1-\sqrt{17}}{4} < X \leq \frac{-1+\sqrt{17}}{4}\right) + P\left(X = \frac{-1+\sqrt{17}}{4}\right) \\ &= F_X\left(\frac{-1+\sqrt{17}}{4}\right) - F_X\left(\frac{-1-\sqrt{17}}{4}\right) + F_X\left(\frac{-1+\sqrt{17}}{4}\right) - \lim_{x \rightarrow \frac{-1+\sqrt{17}}{4}^-} F_X(x) \\ &= \frac{1}{3} \frac{-1+\sqrt{17}}{4} + 1/4 = \frac{1}{6} + \frac{\sqrt{17}}{12} \end{aligned}$$

$$\begin{aligned}
 c) P(Y \leq 1/2) &= P(2X^2 \leq 1/2) = P(0 \leq X \leq 1/4) = P(0 < X \leq 1/4) + P(X=0) \\
 &= F_X(1/4) - F_X(0) + F_X(0) - \lim_{x \rightarrow 0^-} F_X(x) = \frac{1}{3} - \frac{1}{4} + \frac{1}{4} - \frac{1}{4} \\
 &= 1/12
 \end{aligned}$$

$$\begin{aligned}
 d) P(Y > 1/2) &= P(2X^2 > 1/2) = P(X < -1/2 \cup X > 1/2) = P(X < -1/2) + P(X > 1/2) \\
 &= P(X \leq -1/2) - P(X = -1/2) + \{1 - P(X \leq 1/2)\} \\
 &= F_X(-1/2) - \{F_X(-1/2) - \lim_{x \rightarrow -1/2^-} F_X(x)\} + 1 - F_X(1/2) \\
 &= 1/12 - 1/12 + 1/12 + 1 - \frac{5}{12} = 2/3
 \end{aligned}$$

$$\begin{aligned}
 e) P(Y=8) &= P(2X^2=8) = P(X=-2 \cup X=2) = P(X=-2) + P(X=2) \\
 &= F_X(-2) - \lim_{x \rightarrow -2^-} F_X(x) + F_X(2) - \lim_{x \rightarrow 2^-} F_X(x) \\
 &= 0 - 0 + 1 - 1/2 = 1/2
 \end{aligned}$$

3. Gubner Chapter 5, 49

Show that $G(x) = P(X \leq x)$ is left-continuous function

$$G(x) = P(X \leq x)$$

define $A_n = (-\infty, a - \frac{1}{n}]$, $B_n = \{\omega \mid X(\omega) \leq a - \frac{1}{n}\}$

$$\Rightarrow B_n = X^{\leq}(A_n)$$

$$\therefore A_1 \supseteq A_2 \supseteq A_3 \dots \Rightarrow B_1 \supseteq B_2 \supseteq B_3 \dots$$

$$\lim_{n \rightarrow \infty} B_n = \bigcap_{i \in \mathbb{N}} B_n = \{\omega \mid X(\omega) < a\}$$

$$\lim_{n \rightarrow \infty} P(B_n) = P(\lim_{n \rightarrow \infty} B_n) = P(\{\omega \mid X(\omega) < a\}) = P(X < a) = G(a)$$

$$= \lim_{n \rightarrow \infty} P(\{\omega \mid X(\omega) \leq a - \frac{1}{n}\}) = \lim_{x \rightarrow a^-} P(\{\omega \mid X(\omega) \leq x\}) = \lim_{x \rightarrow a^-} G(x)$$

$$\therefore G(a) = \lim_{x \rightarrow a^-} G(x) \text{ ; left continuous function}$$

Show that $P(X=x_0) = G(x_0^+) - G(x_0)$

from $G(a) = \lim_{x \rightarrow a^-} G(x)$

$$G(x_0^+) = \lim_{x \rightarrow x_0^+} G(x), \quad G(x_0) = \lim_{x \rightarrow x_0^-} G(x)$$

$$G(x_0^+) - G(x_0) = \lim_{x \rightarrow x_0^+} G(x) - \lim_{x \rightarrow x_0^-} G(x) = \lim_{x \rightarrow x_0^+} P(X < x) - \lim_{x \rightarrow x_0^-} P(X < x)$$

$$= \lim_{x \rightarrow x_0^+} P(X < x) - G(x_0) \quad \because G(x) \text{ is left continuous function}$$

$$= G(x_0^+) - G(x_0)$$

4. Gubner Chapter 2:5

Y : integer-valued random variable

$$P(Y=n) = P(Y > n-1) - P(Y > n)$$

$$R_Y = \mathbb{N} \quad \therefore P(Y > n) = \sum_{y \in \mathbb{N}, y > n} P(Y=y)$$

$$\Rightarrow P(Y > n-1) - P(Y > n) = \sum_{y \in \mathbb{N}, y > n-1} P(Y=y) - \sum_{y \in \mathbb{N}, y > n} P(Y=y)$$

$$= \sum_{\substack{y \in \mathbb{N}, \\ n-1 < y \leq n}} P(Y=y) = P(Y=n)$$

5. assume $N(w), K(w)$ have the joint pmf $p_{N,K}(n,k) = C/(n^2 k!)$, $n, k \in \mathbb{N}$

a) determine C (Hint: $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$)

$$\sum_{n \in \mathbb{N}} \sum_{k \in \mathbb{N}} p_{N,K}(n,k) = 1$$

$$\Rightarrow C \sum_{n \in \mathbb{N}} \frac{1}{n^2} \sum_{k \in \mathbb{N}} \frac{1}{k!} = C \cdot \frac{\pi^2}{6} \cdot e - 1 = 1 \quad \because e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

$$\therefore C = \frac{6}{\pi^2(e-1)}$$

b) marginal pmfs $p_N(n)$ and $p_K(k)$

$$p_N(n) = \sum_{k \in \mathbb{N}} p_{N,K}(n,k) = \frac{C}{n^2} \sum_{k \in \mathbb{N}} \frac{1}{k!} = \frac{C}{n^2} (e-1)$$

$$= \frac{6}{\pi^2 n^2}$$

$$p_K(k) = \sum_{n \in \mathbb{N}} p_{N,K}(n,k) = \frac{C}{k!} \sum_{n \in \mathbb{N}} \frac{1}{n^2} = \frac{C}{k!} \cdot \frac{\pi^2}{6}$$

$$= \frac{1}{(e-1)k!}$$

c) find probability of $\{\omega \mid |N(\omega) - K(\omega)| = 1\} = A$

$$P(A) = \sum_{n \in \mathbb{N}} p_{N,K}(n, n+1) + p_{N,K}(n+1, n)$$

$$= \sum_{n \in \mathbb{N}} C \left(\frac{1}{n^2(n+1)!} + \frac{1}{(n+1)^2 n!} \right) = \frac{6}{\pi^2(e-1)} \sum_{n \in \mathbb{N}} \left(\frac{1}{n^2(n+1)!} + \frac{1}{(n+1)^2 n!} \right)$$

$$\approx 0.3060$$

6. $p_X(x) = \frac{C}{x^2 + 0.5}, x \in \mathbb{Z}$

a) Find the pmfs of $Y = -X$ and $Z = X$

$$p_X(x) = P(X=x) = \frac{C}{x^2 + 0.5}$$

$$p_Y(y) = P(Y=y) = P(X=-y) = \frac{C}{y^2 + 0.5}, y \in \mathbb{Z}$$

$$p_Z(z) = P(Z=z) = P(X=z) = \frac{C}{z^2 + 0.5}, z \in \mathbb{Z}$$

b) find joint pmfs p_{XY}, p_{XZ}

$$p_{XY}(x,y) = P(X=x, Y=y) = P(X=x, X=-y) = \begin{cases} \frac{C}{x^2 + 0.5} & X = -y \\ 0 & X \neq -y \end{cases}$$

$$p_{XZ}(x,z) = P(X=x, Z=z) = P(X=x, X=z) = \begin{cases} \frac{C}{x^2 + 0.5} & X = z \\ 0 & X \neq z \end{cases}$$

c) joint pmfs shows more relationship between X and Y or X and Z.

7. 52 cards distributed to four persons. 13 cards each

a) Royal Hand

$$P(A) = \frac{(52-12)}{\binom{52}{13}} = \frac{13! \cdot 39!}{52!} \cdot 40$$

b) 40 years (30 normal years and 10 leap years.)

$$= 30 \times 365 + 10 \times 366 = 14610.$$

three times a day : 43830.

$$n = 43830$$

$$p = 6.3 \times 10^{-11}$$

$$\lambda = np = 2.76 \times 10^{-6}$$

A: Royal Hand occurs at least once

B: " " not occurs

$$P(A) = 1 - P(B) = 1 - \binom{n}{0} p^0 (1-p)^n = 1 - (1-p)^n$$

c) $X \sim \text{Pois}(\lambda)$

$$P_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad P_X(0) = e^{-\lambda}$$

$$\lambda = np$$

$$P(A) = 1 - P_X(0) = 1 - e^{-\lambda} = 2.76 e^{-06}$$

8. Gubner Chapter 2-14

m independent auditor, k th auditor processes X_k tax returns per day

$$X_k \sim \text{Pois}(\lambda) \quad P_{X_1, X_2, \dots, X_m}(x_1, x_2, \dots, x_m) = P(X_1=x_1)P(X_2=x_2) \dots P(X_m=x_m)$$

$$P(\text{satisfy}) = P(X_1 > 2)P(X_2 > 2) \dots P(X_m > 2)$$

$$P(\text{unsatisfy}) = 1 - P(\text{satisfy})$$

from $X_k \sim \text{Pois}(\lambda) \quad \lambda > 0$

$$\Rightarrow P(X_k = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\therefore P(X_k > 2) = 1 - P(X_k = 0) - P(X_k = 1)$$

$$= 1 - e^{-\lambda} - \lambda e^{-\lambda}$$

$$= 1 - e^{-\lambda(1+\lambda)}$$

$$\Rightarrow P(\text{unsatisfy}) = 1 - (1 - e^{-\lambda(1+\lambda)})^m$$

$$= (1 - (1 - e^{-\lambda(1+\lambda)})^m)$$

9. Gubner Chapter 2-21

$$X \sim \text{Geometric}(q) = \text{Geo}(1-p)$$

a) Show that $P(X > n) = q^n$

$$P_X(x) = P(X=x) = q(1-q)^{x-1}, \quad q = (1-p)$$

$$\Rightarrow P(X > n) = 1 - \sum_{x=1}^n P(X=x) = 1 - \sum_{x=1}^n q(1-q)^{x-1}$$

$$= 1 - \left(\frac{q(1-q^n)}{1-(1-q)} \right) = 1 - 1 + q^n = q^n$$

b) compute $P(\{X > n+k\} | \{X > n\})$ Hint: if $A \subset B$ then $A \cap B = A$

define $A = \{X > n+k\}$ $B = \{X > n\}$

if $k > 0$ $A \subset B$

$k \leq 0$ $A \supset B$

$$\Rightarrow P(A|B) = \frac{P(A \cap B)}{P(B)} = \begin{cases} P(A)/P(B) & k > 0 \\ P(B)/P(B) = 1 & k \leq 0 \end{cases}$$

$$P(A)/P(B) = \frac{q^{n+k}}{q^n} = q^k = (1-p)^k$$

10. Dobrow 2.28

a) define A: at least one pair people have same birthday

$$P(A) = 1 - P(A^c) = 1 - \left(\frac{365}{365} \cdot \frac{364}{365} \cdots \frac{365-n+1}{365} \right)$$

$$= 1 - \frac{(365)!}{(365-n)! (365)^n}$$

if $n = 23$

$$P(A) = 1 - \frac{(365)!}{(365-23)! (365)^{23}} = 0.51$$

b) $P(A) = 1 - \frac{(365)!}{(365)^n (365-n)!} \sim 0.95$

$$n \sim 47$$

c) A: at least one triplet of people have same birthday.

if $n = 50$ $P(A) = 0.12$

k, number of date two people have same birthday

l, number of date one people have birthday

$$\binom{365}{k+l} \binom{k+l}{l} \frac{n!}{2^k 1^l}$$

$$2k+l=n$$

$$= \binom{365}{n-k} \binom{n-k}{n-k} \frac{n!}{2^k}$$

$$\therefore P(A) = 1 - \sum_{k=0}^{n/2} \frac{\binom{365}{n-k} \binom{n-k}{k} \frac{n!}{2^k}}{(365)^n}$$

$$d) P(A) = 1 - \frac{1}{(365)^n} \sum_{k=0}^{n/2} \binom{365}{n-k} \binom{n-k}{k} \frac{n!}{2^k} \approx 0.5$$

$$\uparrow \quad n \approx 88$$

11.